Null Object

Season: 4 · Enterprise · Category: Enterprise · Level: Intermediate

Intent

- Provide a do-nothing stand-in instead of null.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Mute button that still is a speaker interface.

When to Use

- Microservices / modular monoliths.
- The design force named in Intent shows up more than once.

When Not to Use

- Greenfield CRUD without distribution.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.enterprise.nullobject;

interface Logger {
    void info(String msg);
}

final class ConsoleLogger implements Logger {
    private final StringBuilder sink;
    ConsoleLogger(StringBuilder sink) { this.sink = sink; }
    public void info(String msg) { sink.append(msg).append('\n'); }
}

final class NullLogger implements Logger {
    static final NullLogger INSTANCE = new NullLogger();
    public void info(String msg) { /* no-op */ }
}

/** Prefer Null Object over null checks at call sites. */
public final class OrderService {
    private final Logger logger;

    public OrderService(Logger logger) {
        this.logger = logger == null ? NullLogger.INSTANCE : logger;
    }

    public String place(String id) {
        logger.info("placing " + id);
        return "ok:" + id;
    }
```

Classic Implementation (2)

```
}
```

Modern Java 25 / Functional

- Sealed events + pattern matching for projections.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: enterprise/nullobject/
```

Pitfalls

- Keep demos minimal — production needs idempotency and observability.
- Saga compensations must be careful.

Interview Q&A

- Q: Production concern for Null Object? A: Idempotency, observability, and failure modes.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Provide a do-nothing stand-in instead of null.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Mute button that still is a speaker interface. [0:40] Intent — State the design force in one sentenc...